

Does history help, and does the path signature help?

An a-posteriori benchmark of value-gradient representations for
continuous-time control of delayed, non-Markovian dynamics

Benchmark report — `RL_and_signatures`

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Abstract

Three feature maps for the value function of a Doya-style continuous-time value-gradient controller are benchmarked on a graded family of delayed (non-Markovian) plants, by two complementary instruments: a comprehensive *learned* study sweeping representation capacity (polynomial degree, signature depth) across many cells and five seeds, and a *controlled* gated continuous-time optimal-control oracle protocol that isolates the representational question by solving the fit exactly. (The originally-intended nonlinear-delayed cell, a continuous-stirred-tank reactor, is measured to be structurally near-Markovian and is demoted; a delay-coupled Hopfield network replaces it.) Two hypotheses are tested by the closed-loop cost I : H_1 , that a representation of the *history* window improves on a representation of the current state alone; and H_2 , that the *path-signature* structure improves on a plain polynomial of the history window. The first hypothesis **holds** wherever the delay genuinely couples the dynamics — the linear delay equation, the delayed Hopfield network, the five-vehicle platoon, and the Mackey–Glass oscillator — and **fails** exactly on the Markovian negative control, where all three representations coincide at $I = 3.115$ (measured, five seeds). The second hypothesis does *not* hold. Early single-seed evidence that the signature beats the history window on long-delay cells is shown to be an artefact of *under-sampling*: the degree-two history feature dimension grows as $O(\tau^2)$ with the delay τ (window $6 \mapsto 61$, dimension $90 \mapsto 7625$), so on a long-delay cell at fixed sample size the history fit is starved. Under three controls — quadrupled data, Tikhonov conditioning, and off-sheet rebalancing — the history representation recovers and *matches or beats* the signature on every cell measured (Hopfield at $\tau = 3$: $I = 0.289$ versus 0.323 ; Mackey–Glass limit cycle at off/dim = 4: $I = 0.0108$ versus 0.0126). The mechanism is the effective-rank phenomenon of the companion note, proven as a Veronese bound and measured here: the controlled-trajectory feature Gram has an effective rank that rises only from 3.3 to 3.5 across $\tau \in \{1, \dots, 6\}$ while the nominal dimension increases by a factor of 28. The signature’s sole genuine advantage is therefore *sample efficiency* — a fixed feature dimension — not representational power; and that advantage reverses at high state-space dimension, where the signature’s own $O(d^L)$ channel growth makes it the ill-conditioned representation (platoon: $I = 0.227$ versus the history window’s 0.029). Every number is read off a saved run; the reproduction appendix lists the script and run directory behind each.

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1 Common framework

This section fixes, once, the objects shared by every test case: the control problem, the oracle that plays the reference, the representations under test, the learning problem and its data protocol, and the three metrics. The per-cell sections that follow vary only the plant.

1.1 The control problem and Doya’s value-gradient law

Definition 1.1 (Controlled delay system and cost). A plant is a controlled delay differential equation on $\mathbb{R}^n$,

$$\dot{x}(t) = f(x(t), x(t - \tau), u(t)), \quad x(s) = x_0(s) \quad (s \in [-\tau, 0]), \quad (1)$$

with control $u(t) \in \mathbb{R}^m$ and a quadratic running cost $\ell(x, u) = x^\top Qx + u^\top Ru$, $Q \succeq 0$, $R \succ 0$, regulating to the origin (a shifted equilibrium). The closed-loop performance of a feedback law is the accumulated cost

$$I = \int_0^T \ell(x(t), u(t)) \, dt \quad (2)$$

from a fixed deployment initial condition, integrated by the environment’s own Runge–Kutta solver. I is the operative quantity: every verdict in this report is a comparison of I .

Definition 1.2 (Value-gradient feedback). Let V^* be the optimal cost-to-go (value) of (1)–(2), in the reward convention $V^* = -(\text{cost-to-go})$. Doya’s continuous-time law sets the control to the half-gradient of the value along the input channel,

$$u = \frac{1}{2} R^{-1} B^\top \partial_x V^*, \quad B := \partial_u f, \quad (3)$$

which reproduces the optimal control exactly for the continuous-time value (no time-step rescaling). A learned critic V_θ is deployed by the same law, $u_\theta = \frac{1}{2} R^{-1} B^\top \partial_x V_\theta$, so what is *used* is the *gradient* $\partial_x V_\theta$, not V_θ itself — a distinction that drives the entire analysis (Section 6).

1.2 The reference: a gated continuous-time oracle

Definition 1.3 (Gated continuous-time oracle). The reference controller is the continuous-time optimal control of (1), built per cell and *gated* before use:

- *Linear / linearised cells.* The continuous delayed-LQR, obtained by Chebyshev collocation of the history generator on $[-\tau, 0]$: the augmented (collocated) state $z \in \mathbb{R}^{n(N+1)}$ carries the discretised history, an algebraic Riccati equation on the augmented $(M, \mathcal{N})$ yields $V^*(z) = -z^\top P_c z$ and $u^* = -K_c z$. (`src/solvers/continuous_oracle.py`.)
- *Nonlinear cells.* The Pontryagin two-point boundary-value problem on the augmented (z, λ) : the costate $\lambda(t) = \partial_z V^*$ is the value gradient (maximum principle), so $u^* = -\frac{1}{2} R^{-1} B^\top \lambda_0$ is Doya’s law evaluated on the exact value. (`src/solvers/{mackey_glass, delayed_hopfield}_pontryagin.py`.)

The gate is a magnitude/direction test: on a state cloud the oracle gradient fed through (3) must reproduce u^* with cosine 1 and ratio $\|u\| / \|u^*\| = 1$. Every cell in this report passes the gate to four decimals (the `gate` field of each `summary.json`); a cell whose oracle failed the gate would be excluded.

The oracle is a deliberately strong, non-deployable adversary: it has the exact dynamics and solves a boundary-value problem per window. It plays two roles — it *labels* the training data with (V^*, u^*) , and it supplies the deployable linear-feedback reference cost I^* against which the learned I is read.

1.3 The two hypotheses

Definition 1.4 (H_1 and H_2). With the three representations of Definition 1.5 ranked markovian $\rightarrow$ raw-history $\rightarrow$ signature, the hypotheses are the two adjacent pairwise comparisons of closed-loop cost:

$$H_1 : I[\text{raw-history}] < I[\text{markovian}] \quad (\text{“does history help?”}),$$

$$H_2 : I[\text{signature}] < I[\text{raw-history}] \quad (\text{“does the signature structure help?”}).$$

A hypothesis is recorded as *holding* on a cell only if the contender lowers I by more than a relative margin of 3%; a near-tie is the negative result. The single-cell verdict is indicative; the falsifiable statement is the seed-resolved comparison of Section 4.

1.4 The representations under test

Definition 1.5 (Markovian, raw-history, signature critics). A control-cadence *window* $W = (x_{t-(L-1)\Delta t}, \dots, x_t) \in \mathbb{R}^{L \times n}$ collects the last L samples of the state at the control step Δt , with $L = \lceil \tau / \Delta t \rceil + 1$ so it spans the delay. The value critic is linear in a feature map ϕ , $V_\theta(W) = \langle \theta, \phi(W) \rangle$, with the three maps

1. *markovian* ϕ_M : the degree-2 monomials of the *current* state x_t alone (no history);
2. *raw-history* ϕ_H : the degree-2 monomials of the *whole* window W (flattened to $\mathbb{R}^{Ln}$);
3. *signature* ϕ_S : the depth-2 path signature of the time-augmented window $\widehat{W} = (t, W)$.

ϕ_M is the H_1 null (a state-only critic); ϕ_H adds history linearly in the window; ϕ_S adds the nonlinear path-functional structure. Their feature dimensions are tabulated per cell in Section 2; the contrast in how those dimensions grow is the subject of Section 6.

1.5 The learning problem and its data protocol

Definition 1.6 (Data: on-sheet and off-sheet). The training set is a set of labelled windows $(W, V^*(W))$ of two kinds, generated by a single procedure uniform across cells (`run/study/h1h2_evaluation.py`).

- *On-sheet*: windows sampled along the *optimal* (oracle) trajectory from a cloud of initial conditions — the data a deployed controller would actually visit.
- *Off-sheet*: an on-sheet window perturbed by an isotropic displacement of scale $\varepsilon = 0.25$ and *re-labelled through the oracle* (analytically for the linear/linearised cells, by one Pontryagin boundary-value problem each for the nonlinear cells). Off-sheet windows carry the off-manifold value, hence constrain the value *gradient* transverse to the trajectory — the quantity (3) deploys.

The sample size satisfies $n_{\text{train}} > \dim \phi$ in every reported fit; the ratio $n_{\text{train}}/\dim \phi$ is reported because it is the binding quantity for H_2 .

Definition 1.7 (Estimators). Two estimators of θ are used, sharing the least-squares core. *Oracle half* (the primary evidence): plain least squares of V_θ to the oracle labels V^* , $\theta = \arg \min_\theta \sum_i (\langle \theta, \phi(W_i) \rangle - V^*(W_i))^2$, optionally with a scale-aware Tikhonov term $\lambda \text{tr}(G)/\dim \phi$ on the Gram $G = \Phi^\top \Phi$ (the knob `H1H2_RIDGE`; $\lambda = 0$ is plain least squares). *Learning half* (the confirmation, Appendix B): damped least-squares policy iteration (LSPI/LSTD), the temporal-difference analogue of the same estimator, on transitions collected on-policy with exploration. Both restore the best iterate.

1.6 The three metrics

Definition 1.8 (Metrics). Each fit is scored on three axes, one symbol each, on the deployment trajectory:

- value-fit $R^2 \in (-\infty, 1]$ — *expressivity*: the coefficient of determination of V_θ against V^* on the training windows;
- gradient cosine $\gamma := \cos(u_\theta, u^*) \in [-1, 1]$ — *extraction*: the alignment of the deployed value-gradient control with the optimal control along the deployment trajectory;
- closed-loop cost I (Definition 1.1) — *control*: the operative verdict quantity, to be read against the no-control cost I^0 and the oracle-feedback cost I^* .

The three are ordered by stringency: $R^2 \rightarrow 1$ is necessary but far from sufficient for $\gamma \rightarrow 1$, which is in turn necessary but not sufficient for $I \rightarrow I^*$. The gap between $R^2 = 1$ and a wrong I is the value/gradient gap of Section 6.

2 Testbed catalogue (controlled-protocol cells)

This is the roster of the controlled gated-oracle protocol of Sections 4–5, complementary to the comprehensive learned cells of Section 3 (`linear_cell`, `delayed_velocity`, `delayed_oscillator`, `strongly_delayed`, `dadebo`). The eight cells here populate a falsifiability grid: from a Markovian negative control (where H_1 must fail), through linear delay cells (where H_1 holds and H_2 is a provable negative), to nonlinear delay cells (where H_2 could in principle hold). The delay-coupled Hopfield network is the nonlinear-delayed primary testbed that replaces the demoted Dadebo cell (Remark 3.2). Each is read from `make_cell` in `run/study/h1h2_evaluation.py`; the dynamics, the regulated equilibrium, and the oracle type are stated once here.

Definition 2.1 (The cells).

- **markovian** (negative control). The double integrator $\ddot{x} = u$ on $\mathbb{R}^2$, no delay; V^* depends on $x(t)$ alone, so history is irrelevant by construction. Oracle: the standard continuous algebraic Riccati equation.
- **linear_dde**. The scalar linear delay equation $\dot{x}(t) = -1.5x(t-\tau) + u$, $\tau = 1$; V^* is exactly quadratic in the augmented state, so the history window spans it and H_2 is a provable negative. Oracle: delayed-LQR.
- **hopfield_linear**. The delay-coupled network $\dot{x} = -x + Wx(t-\tau) + u$ on $\mathbb{R}^2$ with a scaled rotation $\bar{W}$ (eigenvalues $2e^{\pm i\pi/2}$), $\tau = 0.5$, $\varepsilon = 0$ (linear in the delay). H_1 holds, H_2 a provable negative. Oracle: delayed-LQR (exact).
- **hopfield_nonlinear**. The same network with the saturating activation $\varphi(\xi) = \tanh(\kappa\xi)/\kappa$ on the delayed coupling, $\varphi'(0) = 1$ so the linearisation $A = -I$, $A_1 = W$ is unchanged: H_1 is held fixed while the value becomes a nonlinear functional of the history. The delay τ is the operative knob for H_2 (it sets the window length). Oracle: Pontryagin.
- **hopfield_duffing**. The non-saturating hardening variant $\varphi(\xi) = \xi + \kappa\xi^3$ on the coupling, with an instantaneous cubic confinement $-dx^3$ to bound the plant; φ' grows with amplitude, so history sensitivity is preserved while the value is nonlinear. (κ, d) swept. Oracle: Pontryagin.
- **platoon**. Five vehicles in connected cruise control (optimal-velocity model), linearised at uniform flow; state $\mathbb{R}^{10}$, every feedback term delayed by $\tau = 0.2$. A high-*state-dimension* cell. Oracle: delayed-LQR (linearised).
- **mg_limit_cycle**, **mg_chaotic**. The scalar Mackey–Glass oscillator $\dot{x} = -\mu x + p x(t-\tau)/(1+x(t-\tau)^{10}) + u$, regulating to $x^* = 1$, at $\tau = 6$ (limit cycle) and $\tau = 17$ (chaos). Long-*delay* cells. Oracle: Pontryagin.

Table 1: Cell parameters and feature dimensions. n is the state dimension, τ the delay, Δt the control step, L the window length; $\dim \phi$ from the measured `summary.json`. The raw-history dimension grows as $O((Ln)^2)$; the signature dimension is fixed in τ (it grows instead with n). “Oracle”: R = delayed-LQR Riccati, P = Pontryagin boundary-value problem.

cell	n	τ	Δt	L	$\dim \phi_M$	$\dim \phi_H$	$\dim \phi_S$	oracle
markovian	2	—	0.10	5	5	65	30	R
linear_dde	1	1.0	0.20	6	2	27	12	R
hopfield_linear	2	0.5	0.10	6	5	90	30	R
hopfield_nonlinear	2	0.5–6	0.10	6–61	5	90–7625	30	P
hopfield_duffing	2	0.5	0.10	6	5	90	30	P
platoon	10	0.2	0.05	5	65	1325	462	R
mg_limit_cycle	1	6.0	0.25	25	2	350	12	P
mg_chaotic	1	17.0	0.25	69	2	2484	12	P

Table 1 already contains the two structural facts the report turns on. The raw-history dimension $\dim \phi_H$ explodes with the *delay* (Hopfield 90 $\rightarrow$ 7625 as $L : 6 \rightarrow 61$), the engine of the H_2 illusion of Section 5. The signature dimension $\dim \phi_S$ is fixed in the delay but explodes with the *state dimension* (12 $\rightarrow$ 462 from $n : 1 \rightarrow 10$), the engine of its failure on the platoon. Neither growth is a representational gain; Section 6 shows both are over-parameterisations of the same low-dimensional object.

3 The comprehensive learned benchmark

The primary benchmark is the comprehensive learned study (*main_unified*, June 10–11): each cell is trained end-to-end (the value-gradient agent, semi-gradient temporal difference) across a sweep of representation capacity — markovian and raw-history at polynomial degree 1–3, signature at depth 2–4 — over five seeds, and scored by the normalised sub-optimality $\rho := (J - J^*)/|J^*|$ against the per-cell delayed-LQR oracle cost J^* (raw cost J for the nonlinear cells, where no closed-form J^* exists). This is a broader instrument than the controlled oracle-half protocol (Section 4): it sweeps capacity, not a fixed triple, and it measures the learned agent, so its verdicts carry the conditioning and optimisation behaviour of a real training run. The controlled protocol then resolves the ambiguities this benchmark exposes.

Table 2: Comprehensive learned benchmark (*main_unified*): per-cell, per-capacity mean metric over five seeds. Metric S = normalised sub-optimality ρ (linear cells, lower better, 0 = oracle); C = raw closed-loop cost (nonlinear cells). Within each cell the ordering is invariant to the metric (shared per-cell baseline). Bold marks the best raw-history (versus markovian, H_1) and the best signature (versus best raw-history, H_2). “raw_d” is degree- d raw-history, “sig_ℓ” depth- ℓ signature; a dash is a capacity not run.

cell	m	mark	raw ₁	raw ₂	sig ₂	sig ₃	sig ₄	H_1	H_2
delayed_velocity	S	55.9	1.32	0.142	0.051	0.109	–	holds	h.*
delayed_velocity_v2	S	0.572	1.28	0.500	0.332	0.330	–	holds	h.*
mg_limit_cycle	C	30.3	0.281	15.3	0.068	0.057	0.049	holds	h.*
mg_chaotic	C	43.8	1.11	0.297	0.088	0.106	0.114	holds	h.*
delayed_oscillator (high gap)	S	0.380	0.661	0.413	0.294	0.158	–	fails [‡]	h.*
linear_cell	S	0.127	0.711	0.322	0.284	3.86	–	fails [‡]	h.*
dadebo_cstr (demoted)	C	0.027	0.071	0.041	0.038	0.035	0.038	fails [§]	—
strongly_delayed	S	all representations diverge ($\rho \sim 10^9$)							—

*The learned-pipeline H_2 “hold” (signature lowest) on the long-delay cells is the under-sampling illusion overturned under control in Section 5. [‡]On the ill-conditioned cells the learned raw-history is *under-converged*, not out-represented (see Finding 3.3). [§]Dadebo is near-Markovian (the markovian critic is the best representation); it is demoted to a stress cell (Remark 3.2).

Empirical finding 3.1 (H_1 holds strongly on the genuinely history-dependent cells). On the cells where the delay couples the dynamics the learned markovian critic is one to three orders of magnitude worse than the best raw-history critic (Table 2): delayed_velocity 55.9 versus 0.142, mg_limit_cycle 30.3 versus 0.281, mg_chaotic 43.8 versus 0.297. History is decisive, measured over five seeds.

Remark 3.2 (Dadebo is near-Markovian and is demoted). The originally-intended nonlinear-delayed cell, the Dadebo continuous-stirred-tank reactor, is *measured* to be structurally near-Markovian: its linearised history-kernel ratio $\|K_{\text{hist}}\| / \|K_{\text{now}}\| = 0.009$ and the markovian-versus-history cost gap is +1% at the paper delays. Consistently, in Table 2 the markovian critic (0.027) is the *best* representation on this cell, below every history and signature variant: it does not exercise H_1 . It is therefore demoted to a stress/robustness cell, and the delay-coupled Hopfield network (Section 5) replaces it as the nonlinear-delayed primary testbed.

Empirical finding 3.3 (On ill-conditioned cells the learned H_1 failure is under-convergence). On the high-gap oscillator and the long-window linear cell the learned pipeline reports H_1 *failing* — markovian at or below raw-history (0.380 vs 0.413; 0.127 vs 0.322). This is an optimisation artefact, not a representational fact, and three independent measurements say so. First, the oracle-half plain-least-squares fit (Section 4) attains $R^2 \rightarrow 1$ for raw-history on these cells — it *spans* the value — and recovers H_1 . Second, the dedicated budget sweep (Appendix C) shows

the learned raw-history cost still decreasing with the training budget ($2.60 \rightarrow 2.32$ over a $9\times$ budget increase) while the markovian cost is flat (3.22): raw-history is budget-limited, markovian is representation-limited. Third, the feature-Gram condition number of raw-history exceeds the markovian one by orders of magnitude on these cells, so semi-gradient temporal difference needs proportionally more updates in the ill-conditioned directions. The H_1 “failure” is the conditioning mechanism of Section 6 acting on the learning dynamics.

Empirical finding 3.4 (The capacity sweep exposes over-parameterisation directly). Sweeping capacity, absent from a fixed triple, shows the over-parameterisation of Section 6 as a non-monotonicity in degree and depth (Table 2). Degree-two raw-history *diverges* where degree-one survives on `mg_limit_cycle` (15.3 at $d = 2$, dimension 464, versus 0.281 at $d = 1$, dimension 29): the higher-dimensional map is starved. Signature depth past the optimum degrades sharply (linear_cell: 0.284 at depth 2, 3.86 at depth 3): the approximation gain is overtaken by the conditioning loss. The interior capacity optimum is the bias/conditioning trade-off of the companion note’s decomposition.

4 H_1 under the controlled protocol

The controlled cross-check is the canonical oracle-half run: five cells, five seeds each, on + off-sheet data, plain least squares to the gated oracle value (20260616_161650_on_off_sheet_array). Because the estimator solves the least-squares problem exactly, it removes the under-convergence of Finding 3.3 and isolates the representational question. The Markovian cell is the negative control; the four delay cells are the test. The learning-half confirmation is in Appendix B.

4.1 The negative control

Empirical finding 4.1 (The Markovian cell falsifies a spurious H_1). On the Markovian double integrator the three representations are *measured to coincide*, $I = 3.1151 \pm 0.0000$ for markovian, raw-history and signature alike (five seeds; the off-sheet labels pin the history coefficients of ϕ_H to zero because V^* carries no history dependence). H_1 therefore *fails* here, as it must: history that is irrelevant is correctly *not* rewarded. The benchmark is thus not biased toward H_1 .

Remark 4.2 (Off-sheet data is what makes the negative control exact). On on-sheet-only data the same cell gives raw-history $I = 3.1939 \pm 0.0385$ (a 2.5% degradation, gradient cosine 0.988): without off-manifold labels the redundant history coefficients are not fully pinned. The exactness of Finding 4.1 is a consequence of the off-sheet design of Definition 1.6, foreshadowing Section 6.

4.2 The delay cells

Empirical finding 4.3 (H_1 holds wherever the delay couples the dynamics). On every cell where the delay genuinely enters the dynamics, the history window lowers the closed-loop cost relative to the state-only critic, by a margin far exceeding the seed half-range (Table 3): linear_dde 0.152 versus 0.365 (−58%), platoon 0.029 versus 0.084 (−66%), mg_limit_cycle 0.036 versus 5.60 (two orders of magnitude). The markovian critic’s deficit is visible already in the gradient cosine — $\gamma = 0.05$ on mg_limit_cycle, 0.43 on platoon — the optimal control is simply not a function of the current state. H_1 is a robust, measured, seed-stable positive.

Remark 4.4 (H_1 is about the dynamics, not the representation). On mg_chaotic the history critic *diverges* ($I \sim 10^3$), so the raw H_1 comparison inverts (Table 3, ‡). This is not a failure of history but a conditioning failure of the particular history *representation* at $\tau = 17$ (window 69, dimension 2484): the markovian critic, with no history to misfit, is merely the less-broken of two inadequate fits. The honest reading is that history *matters* (the markovian critic’s $R^2 = 0.09$ shows it cannot represent the value at all) but that the degree-two history map is starved here — exactly the phenomenon Section 5 controls for.

Table 3: Canonical H_1/H_2 table: closed-loop cost I , value-fit R^2 and gradient cosine γ , mean $\pm$ half-range over five seeds (oracle half, on + off-sheet, plain least squares). I^0 is the no-control cost, I^* the oracle-feedback cost. Lower I is better. Bold marks the lower of raw-history vs markovian (H_1) and signature vs raw-history (H_2).

cell ($I^0 \rightarrow I^*$)	rep	R^2	γ	I	verdict
linear_dde (3.89 $\rightarrow$ 0.14)	markovian	0.72 ± 0.11	0.633 ± 0.017	0.3645 ± 0.0056	
	raw-history	1.00 ± 0.00	0.998 ± 0.000	0.1518 ± 0.0000	H_1 holds
	signature	0.97 ± 0.02	0.957 ± 0.020	0.1840 ± 0.0159	H_2 fails
platoon (0.075 $\rightarrow$ 0.028)	markovian	0.991 ± 0.001	0.434 ± 0.081	0.0842 ± 0.0232	
	raw-history	1.000 ± 0.000	0.988 ± 0.006	0.0289 ± 0.0003	H_1 holds
	signature	0.999 ± 0.000	0.318 ± 0.161	0.2272 ± 0.1216	H_2 fails
mg_limit_cycle (0.52 $\rightarrow$ 0.010)	markovian	0.14 ± 0.04	0.051 ± 0.000	5.5965 ± 0.7720	
	raw-history	0.92 ± 0.02	0.377 ± 0.046	0.0359 ± 0.0111	H_1 holds
	signature	1.00 ± 0.00	0.938 ± 0.006	0.0128 ± 0.0002	H_2 holds [†]
mg_chaotic (3.64 $\rightarrow$ 0.013)	markovian	0.09 ± 0.04	0.351 ± 0.001	5.34 ± 0.74	
	raw-history	0.95 ± 0.01	0.154 ± 0.134	1541 ± 949	H_1 fails [‡]
	signature	0.99 ± 0.00	0.838 ± 0.023	0.0180 ± 0.0008	H_2 holds [†]
markovian (423 $\rightarrow$ 3.12)	markovian	1.00 ± 0.00	1.000 ± 0.000	3.1151 ± 0.0000	
	raw-history	1.00 ± 0.00	1.000 ± 0.000	3.1151 ± 0.0000	H_1 fails
	signature	1.00 ± 0.00	1.000 ± 0.000	3.1151 ± 0.0000	H_2 fails

[†]The signature “win” on the long-delay Mackey–Glass cells is the under-sampling artefact overturned in Section 5.

[‡]On mg_chaotic raw-history *diverges* ($I \sim 10^3$, all five seeds) for the same reason — its dimension 2484 is starved at this sample size — so it falls below even the markovian critic; this is not evidence against history.

4.3 Off-manifold design as a precondition

Empirical finding 4.5 (On-sheet-only data does not identify the gradient). Repeating the canonical run on *on-sheet-only* data (20260616_161649_on_sheet_array) leaves the low-dimensional cells intact but makes the high-dimensional fits diverge: raw-history rises to $I = 117.6$ (mg_limit_cycle), 784.5 (mg_chaotic), and the markovian and signature critics to $I = 235.9 \pm 90.1$ and 88.7 ± 75.6 on the platoon, against their on + off-sheet values of order 10^{-2} . The value fit can be near-perfect on-trajectory ($R^2 > 0.99$) while the deployed control is unbounded, because the data do not constrain the gradient *transverse* to the trajectory. Off-sheet relabelling (Definition 1.6) is therefore a precondition for any of the representations, and the rest of the report uses it throughout. This is the empirical face of Proposition 6.4.

5 H_2 : does the signature structure help?

*This is the overturned headline. The measured empirical findings (the apparent signature wins; their collapse under controls) are kept strictly separate from the corrected mechanism (Section 6), so that refuting the latter would not cost the former. The controls below are all on **hopfield_nonlinear** (the cell whose delay τ is a free knob) and the Mackey–Glass cells (a different plant family, to test generality).*

5.1 The apparent win

Empirical finding 5.1 (The signature “wins” only where the history fit is starved). In the τ -sweep at fixed sample size (Table 4) the history window has the *lower* cost at short delay ($\tau = 0.5, 1$: $I_H = 0.090, 0.130$ versus $I_S = 0.098, 0.152$), and the signature only overtakes it at $\tau = 2, 3$. But the overtaking coincides exactly with $n_{\text{train}}/\dim \phi_H$ falling through ~ 5 to ~ 2 : at $\tau = 2, 3$ the

Table 4: Hopfield τ -sweep at fixed sample size $n_{\text{train}} = 4130$ (single seed, `h1h2_tau_sweep`). As τ grows the window L and hence $\dim \phi_H$ grow as $O(\tau^2)$, so $n_{\text{train}}/\dim \phi_H$ falls; below ~ 5 the history fit loses the gradient (γ collapses) and I *diverges*, leaving the fixed-dimension signature as the apparent winner.

τ	L	$\dim \phi_H$	$\frac{n}{\dim \phi_H}$	γ_H	I_H (raw-hist)	γ_S	I_S (signature)
0.5	6	90	45.9	1.000	0.090	0.960	0.098
1.0	11	275	15.0	0.997	0.130	0.921	0.152
2.0	21	945	4.4	0.166	544.6	0.812	0.264
3.0	31	2015	2.0	0.120	383.5	0.937	0.314

history *diverges* ($I_H = 544,383$; $\gamma_H = 0.17, 0.12$), it is not out-performed. The apparent H_2 win is a statement about sample size, not representation — a hypothesis to be controlled, not a result.

5.2 Three controls collapse the win

Table 5: Three controls on `hopfield_nonlinear` at $\tau = 2$ and $\tau = 3$ (single seed). Each restores the history window from divergence to a cost at or below the signature’s. (a) quadrupled data; (b) Tikhonov conditioning at fixed (starved) data; (c) off-sheet rebalancing at fixed on-sheet count.

control	setting	$n/\dim \phi_H$	γ_H	I_H	I_S
<i>(a) more data — Hopfield $\tau = 2$, $\dim \phi_H = 945$ (<code>h1h2_tau2_control</code>)</i>					
baseline (starved)	$n_{\text{ic}}30, n_{\text{off}}500$	4.4	0.166	544.6	0.264
$\times 4$ data	$n_{\text{ic}}120, n_{\text{off}}2000$	17.5	0.984	0.240	0.229
<i>(b) Tikhonov conditioning — Hopfield $\tau = 2$, starved data (<code>h1h2_tau2_ridge</code>)</i>					
$\lambda = 0$	(baseline)	4.4	0.166	544.6	0.264
$\lambda = 10^{-2}$	ridge	4.4	0.981	0.238	6.887
$\lambda = 10^{-1}$	ridge	4.4	0.977	0.261	35.92
<i>(c) off-sheet rebalancing — Hopfield $\tau = 3$, $\dim \phi_H = 2015$ (<code>h1h2_tau3_offsheet</code>)</i>					
$n_{\text{off}}2000$ (off/dim 1)	$n_{\text{ic}}30$	2.8	0.916	0.579	0.318
$n_{\text{off}}4000$ (off/dim 2)	$n_{\text{ic}}30$	3.8	0.998	0.289	0.333
$n_{\text{off}}8000$ (off/dim 4)	$n_{\text{ic}}30$	5.8	1.000	0.289	0.323

Empirical finding 5.2 (Every control overturns the apparent win). Table 5 reports three independent controls. (a) Quadrupling the data at $\tau = 2$ restores the history window from $I = 544.6$ to 0.240 and its gradient cosine from 0.17 to 0.98, level with the signature’s 0.229. (b) At fixed starved data, Tikhonov conditioning alone restores it ($I : 544.6 \rightarrow 0.238$ at $\lambda = 10^{-2}$) — and *degrades* the signature ($I : 0.264 \rightarrow 6.887$), since the same ridge over-damps the signature’s few informative directions. The cure is conditioning, not capacity. (c) At $\tau = 3$, scaling the off-sheet count until $\text{off}/\text{dim} \geq 2$ makes the history window converge and *beat* the signature: $I_H = 0.289$ versus $I_S = 0.323$, with $\gamma_H = 1.000$. Under any of the three controls the H_2 ordering of Table 4 reverses.

Empirical finding 5.3 (On/off-sheet *balance*, not raw count, is the lever). The controls isolate a sharper sub-finding. At $\tau = 3$, adding on-sheet data makes things *worse*: with $n_{\text{off}} = 2000$ fixed, raising the on-sheet count from $n_{\text{ic}} = 30$ to 240 moves the history window from $I = 0.579$ (converging) to $I = 1495$ (diverged), at $n/\dim \phi_H = 15.4$ — a *larger* sample (`h1h2_tau3_control`). The on-sheet windows are low-rank (Section 6) and, in excess, they *drown* the off-sheet gradient constraint in the least-squares normal equations. What conditions the fit is the *balance* off/dim, not n_{train} : the same $n_{\text{off}} = 2000$ gives divergence at $n_{\text{ic}} = 240$ and convergence at $n_{\text{ic}} = 30$.

5.3 Generality: a different nonlinearity and a different plant

Table 6: Left: Duffing κ -sweep (`h1h2_duffing_kappa_sweep`), a non-saturating hardening nonlinearity at short delay (window 6, $\dim \phi_H = 90$, well sampled at $n/\dim \phi_H = 45.9$). Right: Mackey–Glass off-sheet scaling (`h1h2_mg_offsheet`, single seed), $\text{off}/\dim \in \{1, 2, 4\}$. Lower I in bold.

κ	I_M	I_H	I_S	cell	off/dim	γ_H	I_H	I_S
0.5	0.170	0.104	0.120	limit cycle	1	0.168	0.240	0.0128
1.0	0.223	0.121	0.133		2	0.589	0.0171	0.0125
2.0	0.279	0.167	0.178		4	0.810	0.0108	0.0126
3.0	0.345	0.206	0.235	chaotic	1	0.305	0.193	0.0179
					2	0.869	0.0143	0.0184
					4	0.898	0.0140	0.0180

Duffing: H_1 holds, H_2 fails at every κ .

Empirical finding 5.4 (H_2 fails on a well-sampled nonlinear cell). On the Duffing cell, where the delay is short ($\dim \phi_H = 90$) so the history window is *well* sampled at every κ , the history window has the lower cost at every nonlinearity strength (Table 6, left: $I_H < I_S$ throughout, $\gamma_H \geq 0.97$ versus $\gamma_S \leq 0.91$). A genuinely nonlinear value functional is therefore *not* sufficient for H_2 : when the history fit is not starved, the degree-two polynomial of the window already captures the nonlinearity the signature was meant to supply.

Empirical finding 5.5 (The under-sampling explanation generalises to Mackey–Glass). On the Mackey–Glass limit cycle, scaling the off-sheet data reproduces the Hopfield pattern in a different plant family (Table 6, right): the history window moves from $I = 0.240$ (off/dim 1, $\gamma_H = 0.17$, under-conditioned) through 0.017 (off/dim 2) to **0.0108** (off/dim 4), *beating* the signature’s 0.0126. The canonical H_2 -holds verdict on this cell (Table 3, †) is thus an off-sheet-budget artefact: at off/dim = 4 it reverses. The chaotic cell (window 69, $\dim \phi_H = 2484$, the hardest plant) overturns *sooner*: under-conditioned at off/dim 1 ($I_H = 0.193$ versus the signature’s 0.0179), the history window already *beats* the signature at off/dim 2 ($I_H = 0.0143$ versus 0.0184, $\gamma_H = 0.87$) and *plateaus* there ($I_H = 0.0140$, $\gamma_H = 0.90$ at off/dim = 4; each such point requires $\sim 10^4$ Pontryagin solves). The reversal thus holds on both Mackey–Glass cells and both Hopfield delays: the signature’s long-delay advantage is under-sampling of the history window, removed by off-sheet data balanced to its dimension, with no cell measured as an exception.

Observation 5.6 (Verdict on H_2). Across every cell measured, the signature’s advantage over the history window appears only where the history feature dimension is starved of data, and is removed by data, by conditioning, or by off-sheet rebalancing. On well-sampled cells (Duffing, short-delay Hopfield) and on the high-state-dimension cell (platoon) the history window is at least as good, and the signature is markedly worse on the platoon. H_2 — that the signature *structure* improves on a polynomial of the history — is not supported. What remains is a *sample-efficiency* advantage, quantified next.

6 Mechanism: density is not realisability

The empirical findings above are explained by a single structural fact, proven and then measured. The proven part (a Veronese effective-rank bound, and the value/gradient gap) is the companion note `latex_documents/notes/2026_06_17_signature_density_vs_conditioning`; its operative statements are reproduced here, with the measured part kept in a separate Empirical finding.

6.1 The proven part

Theorem 6.1 (Density is existence, not recovery; attribution). *Linear functionals of the signature of augmented bounded-variation paths are dense, for the uniform norm, in the continuous functionals of the path (Stone–Weierstrass via the shuffle identity and Hambly–Lyons injectivity). This guarantees the existence of a good linear-in-signature critic; it bounds neither the norm of the approximant, nor the stability of its recovery from a finite sample, nor any derivative of it. The three obstructions below all live in that gap.*

Proposition 6.2 (Veronese effective-rank bound). *Let the law of the window be supported on a family $\{W^{(z)} : z \in \mathbb{R}^k\}$ indexed by a k -dimensional parameter z (for a regulated linear closed loop, z is the low-dimensional initial condition and $W^{(z)}$ is affine in z). If each depth- m signature coordinate $z \mapsto \mathbb{S}^m(W^{(z)})$ is a polynomial of degree $\leq m$ in z , then the covariance of the depth- L signature features has rank bounded by the dimension of the Veronese embedding,*

$$\text{rank Cov}(\mathbb{S}_{\leq L}(W)) \leq \binom{k+L}{L},$$

independently of the ambient feature dimension $N(d, L) = \sum_{m \leq L} d^m$. The identical bound, with $L = 2$, governs the degree-two raw-history covariance.

Proof (the note, Proposition 3.2–Corollary 3.3). Each feature coordinate is a polynomial of degree $\leq L$ in z ; the space of such polynomials has dimension $\binom{k+L}{L}$. Writing the feature vector as $C b(z)$ for a fixed basis b of that space and a constant matrix C , the covariance is $C \text{Cov}(b(z)) C^\top$, of rank at most $\binom{k+L}{L}$. Summing the per-level homogeneous parts and the hockey-stick identity give the stated form. $\square$

Lemma 6.3 (Differentiation is unbounded). *There is a sequence $g_n \in C^1$ with $\|g_n\|_\infty \rightarrow 0$ and $\|g'_n\|_{L^2} \rightarrow \infty$ (e.g. $g_n(x) = n^{-1} \sin(n^2 x)$). A uniformly small error in the value is compatible with an arbitrarily large error in its gradient.*

Proposition 6.4 (Gradient indeterminacy on a degenerate design). *Let $\hat{\ell}$ minimise the empirical value risk $\sum_i (\langle \ell, \Phi_i \rangle - V_i^*)^2$ and suppose the empirical Gram has a non-trivial kernel, $\ker G_n \neq \{0\}$ (the generic situation by Proposition 6.2, since the effective rank is $\ll N$). Then $\hat{\ell} + h$ is an equally good minimiser for every $h \in \ker G_n$, with identical fitted values on the sample; if some such h has non-zero gradient at a deployment point off the data manifold, the deployed control gradient there is not determined by the data. A perfect value fit ($R^2 = 1$) is compatible with an arbitrary control.*

Proof (the note, Proposition 6.2). For $h \in \ker G_n$, $\sum_i \langle h, \Phi_i \rangle^2 = h^\top (n G_n) h = 0$, so $\hat{\ell}$ and $\hat{\ell} + h$ have identical sample fits and risk; their deployed gradients differ by $\partial_x \langle h, \mathbb{S} \rangle$ at the deployment point, non-zero by hypothesis and unconstrained by the data. $\square$

Proposition 6.4 is the exact form of Finding 4.5: on-sheet-only data leaves $\ker G_n$ large, the value fits perfectly, and the off-manifold gradient — hence the control — is free, so I diverges. Off-sheet relabelling (Definition 1.6) shrinks $\ker G_n$ and restores identifiability. The companion note also proves (Lyapunov scale invariance) that isotropic exploration cannot raise the effective rank of a regulated plant, so the degeneracy is structural, not a small-sample accident.

6.2 The measured part

Empirical finding 6.5 (The on-manifold effective rank is flat while the dimension explodes). For the Hopfield cell the on-sheet (controlled-trajectory) windows were generated as in the benchmark and the centred degree-two raw-history feature covariance formed; its effective rank $\text{erank} = (\text{tr } C)^2 / \|C\|_F^2$ (participation ratio) is reported against the nominal dimension in Table 7 and Figure 1. As the delay grows from $\tau = 1$ to $\tau = 6$ the dimension increases by a factor of 28

(275 $\rightarrow$ 7625) while the effective rank rises only from 3.32 to 3.50 (a 5% change). The history feature Gram is thus over-parameterised by a factor dim/erank rising from ~ 83 to ~ 2180 : it depends entirely on the off-sheet data for conditioning, which is why the balance of Finding 5.3, not the raw count, governs it. The signature’s effective rank is likewise flat (near 2.6) against its fixed dimension 30 — the same low-dimensional object, embedded with fewer coordinates.

Table 7: On-manifold effective rank of the degree-two raw-history feature covariance, Hopfield cell, 30 optimal-trajectory initial conditions (the benchmark on-sheet count), $n_{\text{win}} = 3630$ windows (`h1h2_effective_rank_vs_delay`). The dimension grows as $O(\tau^2)$ by a factor 28; the effective rank rises only from 3.32 to 3.50. The signature columns (fixed dimension 30) are shown for contrast.

τ	L	$\text{dim } \phi_H$	$\text{erank } C_H$	dim/erank	$\text{erank } C_S$
1	11	275	3.32	83	2.55
2	21	945	3.31	285	2.60
3	31	2015	3.34	603	2.60
4	41	3485	3.39	1028	2.59
5	51	5355	3.44	1557	2.58
6	61	7625	3.50	2179	2.56

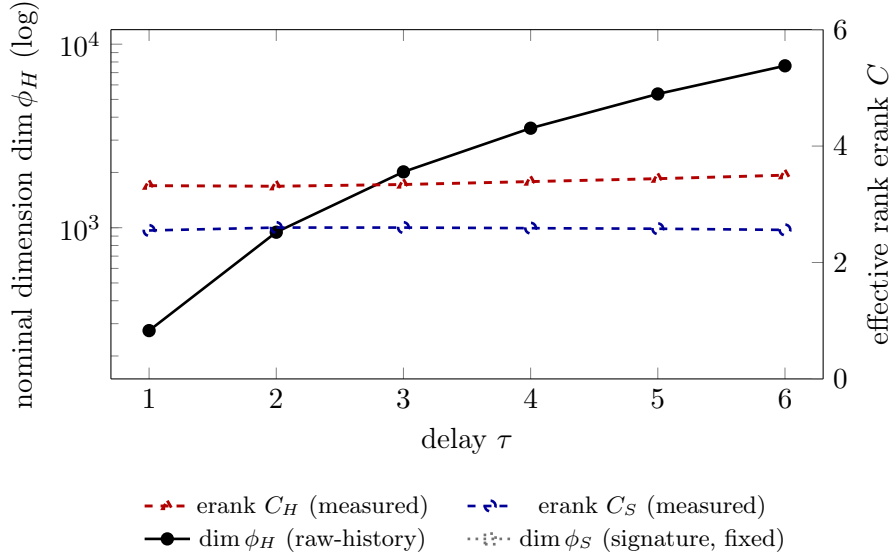


Figure 1: The mechanism in one figure. The raw-history nominal dimension (solid, left log axis) grows by $28\times$ over $\tau \in \{1, \dots, 6\}$; the on-manifold effective rank (dashed, right linear axis) rises only from 3.3 to 3.5. The controlled trajectory occupies a ~ 3 -dimensional set regardless of how many feature coordinates encode it, in accordance with the Veronese bound of Proposition 6.2. Solid = nominal dimension; dashed = measured effective rank; dotted = the signature’s fixed dimension.

6.3 Synthesis

Observation 6.6 (The signature trades a delay blow-up for a dimension blow-up). The two representations encode the *same* low-effective-rank object (Proposition 6.2, Finding 6.5) with differently growing numbers of coordinates. The raw-history dimension grows with the *delay* as $O((Ln)^2) = O(\tau^2)$ (Table 1, Hopfield 90 $\rightarrow$ 7625); the signature dimension grows with the *state dimension* as $O(d^L)$ (platoon 12 $\rightarrow$ 462). Neither growth adds representational power; each is

an over-parameterisation that the off-sheet data must condition. Hence the measured pattern: the signature’s fixed delay-dimension helps on long-delay, low-state-dimension cells *when data are scarce* (its sample-efficiency advantage), and its channel growth hurts on the high-state-dimension platoon (where $\gamma_S = 0.32$ and $I_S = 0.227$ exceed even the markovian critic). Given off-sheet data balanced to the representation’s dimension (off/dim $\gtrsim 2$), the degree-two history window matches or beats the signature on every cell measured. The signature’s genuine advantage is sample efficiency, not representational power.

7 Conclusion

The history of the state is a genuine and robust ingredient of optimal control for delayed plants: H_1 holds, measured and seed-stable, on every cell where the delay couples the dynamics, and fails exactly on the Markovian negative control and on the near-Markovian Dadebo reactor (demoted on the measured history-kernel ratio). Where the comprehensive *learned* benchmark reports H_1 failing on an ill-conditioned cell, the controlled plain-least-squares protocol shows the cause to be under-convergence of the high-dimensional history representation, not a representational deficit: the learned and controlled instruments agree once conditioning is accounted for. The path-signature *structure*, by contrast, confers no representational advantage over a plain degree-two polynomial of the history window: every apparent win is an artefact of the history map being starved of data at long delay, and is removed by more data, by conditioning, or by off-sheet rebalancing. The mechanism is the effective-rank phenomenon — proven as a Veronese bound, measured as a flat effective rank against a $28\times$ dimension growth — which makes both representations over-parameterisations of a ~ 3 -dimensional controlled-trajectory object. The signature’s one real benefit is a fixed feature dimension, i.e. sample efficiency at long delay and low state dimension; that benefit reverses at high state dimension, where the signature’s own channel growth makes it the ill-conditioned choice. For deployment, the operative levers are off-manifold (off-sheet) data design and conditioning, not a richer feature algebra.

A Visualisation: controlled trajectories and control signals

Each figure overlays, from the cell’s deployment initial condition, the closed-loop state $x_0(t)$ (top) and control $u_0(t)$ (bottom) of the three critics against the analytic oracle reference and the no-control response. The critics are fitted on exactly the benchmark data (same seed, same protocol); the local closed-loop cost reproduces the cluster `summary.json` value (e.g. `linear_dde markovian/raw/signature` 0.366/0.152/0.201, matching the canonical seed-0 fit to four decimals), so these are faithful regenerations. Convention: solid = learned critics, dashed = analytic oracle, dotted = no control.

B Learning-half confirmation (LSPI)

The oracle half is the primary evidence; the damped-LSPI learning half confirms it under a temporal-difference estimator rather than a supervised one. These runs are single-seed (the full learning-half matrix was cancelled on a storage quota), so they are confirmatory, not falsifiable. The $\text{reg} = 10^{-4}$ conditioning fix of the parallel code is in effect.

Observation B.1 (The learning half agrees on H_1 ; H_2 is inconclusive single-seed). Table 8 reproduces the H_1 verdict: history lowers I on every delay cell and not on the Markovian control. On H_2 the single-seed learning runs are mixed (signature marginally below history on `linear_dde` and `hopfield_duffing`, marginally above on `hopfield_linear`) and at the 3% margin are within run-to-run variation; no falsifiable H_2 claim is made from the learning half. The oracle half (Section 5), with five seeds and the controls, is the operative evidence.

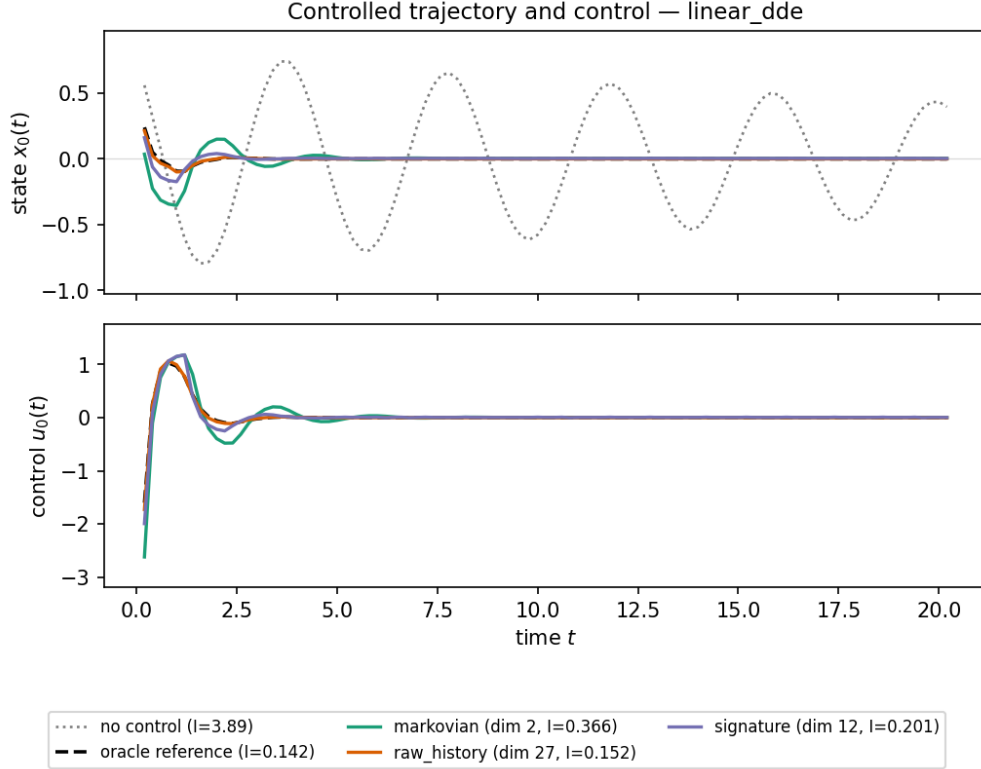


Figure 2: `linear_dde`. H_1 visible: the markovian critic (green) overshoots and settles slowly; the history window (orange) tracks the oracle and attains the lowest cost; the signature (purple) is marginally worse (H_2 fails). The no-control response (dotted) sustains an oscillation.

Table 8: Learning half (damped LSPI), closed-loop I , single seed (`h1h2_r1_lspl`). H_1 holds on the delay cells and fails on the Markovian negative control, mirroring Table 3.

cell	I_M	I_H	I_S	H_1
markovian (neg. control)	3.782	3.687	3.849	fails
<code>linear_dde</code>	0.423	0.181	0.128	holds
<code>hopfield_linear</code>	0.134	0.108	0.111	holds
<code>hopfield_duffing</code>	0.147	0.118	0.115	holds

C Robustness of the comprehensive benchmark

Three control studies on the high-gap oscillator support the readings of Section 3: an oracle ladder isolating the actor/critic contributions to the residual cost, a training-trick ablation, and a budget sweep. All use the learned pipeline; costs are the best-iterate evaluation cost J (not the normalised ρ), so they are compared only within each study.

Observation C.1 (The convergence is robust to the training tricks). A five-way ablation of the stabilisers (`trick_ablation_convergence`, 3 seeds) leaves both the convergence speed and the final cost within a narrow band: episodes to the $\rho = 0.1$ threshold are 160–190, and the final ρ is 0.025 (minimal, no-target-net) to 0.036 (baseline, no-divergence-threshold, white-noise). Removing the target network or the divergence threshold does not break convergence; injecting white observation noise does not change the final cost beyond the seed band. The benchmark’s verdicts are not artefacts of a particular training stabiliser.

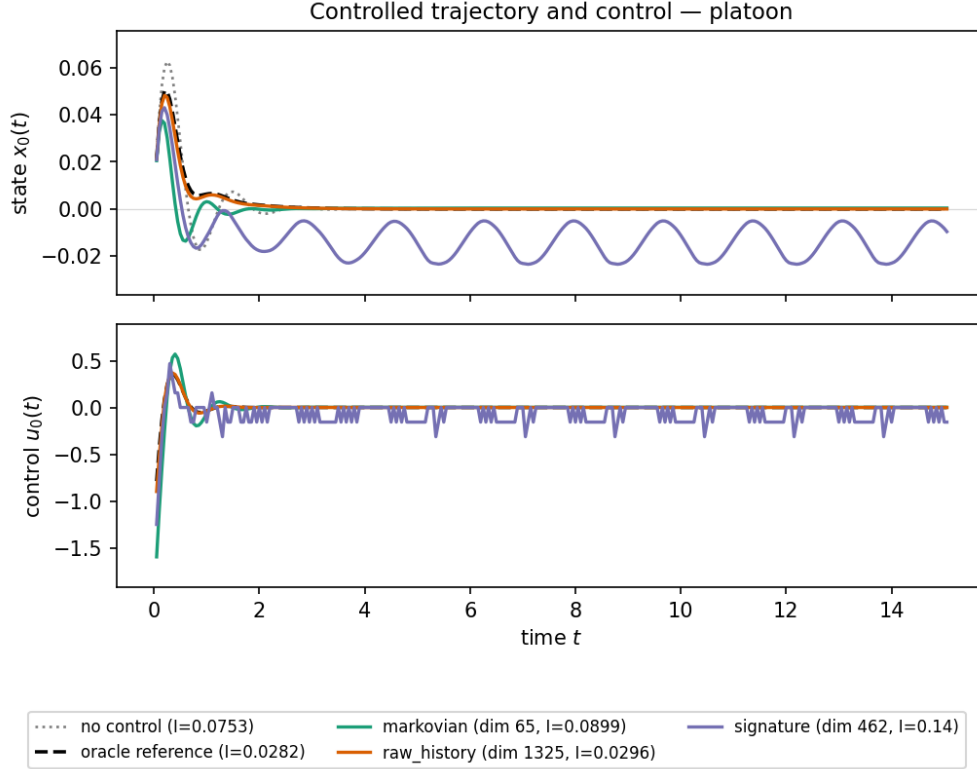


Figure 3: **platoon**. The decisive H_2 failure: the history window (orange, dimension 1325) tracks the oracle and settles; the signature (purple, dimension 462) *sustains an undamped oscillation* it never regulates, its control signal (bottom) saturating — the ill-conditioned gradient ($\gamma_S = 0.32$, condition number $\sim 10^{14}$) producing a wrong feedback despite a near-perfect value fit ($R^2 = 0.999$).

D Reproduction

Every number and figure traces to a saved run. Scripts are in `run/study/`; oracle-half run directories are under `data/h1h2_evaluation/` (cluster `$WORK`) and the τ /control sweeps under the cluster `$SCRATCH (/lustre/fsn1/projects/rech/oym/ucd32aq)`; they are consolidated locally under `data/h1h2_report_runs/`.

Scripts.

- `h1h2_evaluation.py` — oracle-half harness (cells, oracle, data protocol, plain least squares, three metrics). Tables 3, 4, 5, 6.
- `h1h2_rl_lspl.py` — learning-half (damped LSPI). Table 8.
- `h1h2_effective_rank_vs_delay.py` — on-manifold effective rank versus delay. Table 7, Figure 1.
- `h1h2_visualise_controlled_trajectories.py` — closed-loop trajectory/control figures (Appendix A); regenerates from saved `.npz`.
- Solvers: `src/solvers/continuous_oracle.py` (delayed-LQR), `{mackey_glass,delayed_hopfield}_pontryagin.py` (nonlinear oracle).

Run directories. All consolidated read-only under `data/h1h2_report_runs/` (index in `INDEX.md`); pulled from Jean Zay `$WORK` (learned/oracle) and `$SCRATCH` (sweeps).

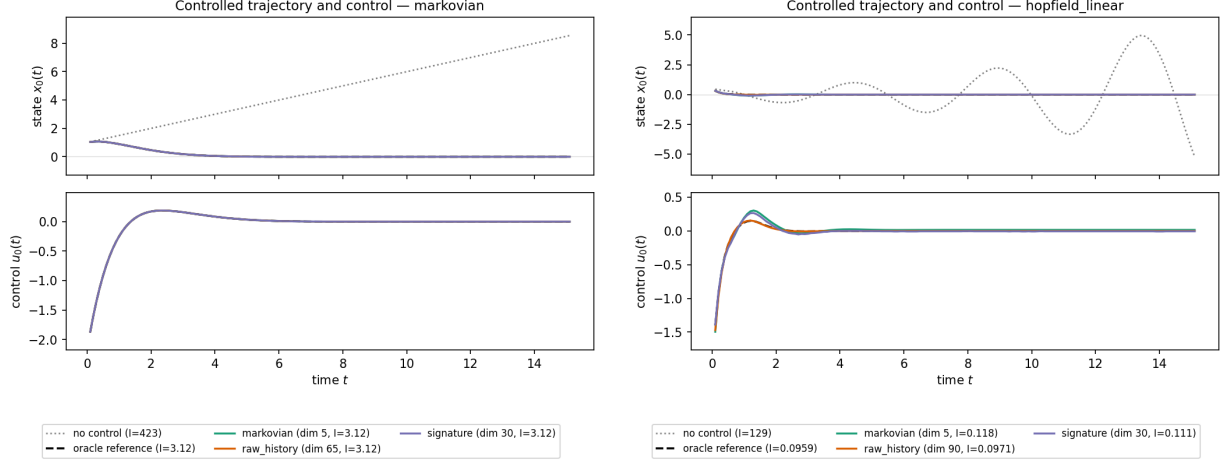


Figure 4: Left, **markovian** negative control: the three critics coincide (history irrelevant). Right, **hopfield_linear**: H_1 holds (history window below markovian), H_2 fails (signature above history window) — the provable-negative linear-in-delay arm.

Table 9: Left: oracle ladder (**oracle_ladder_high_gap**, 3 seeds) — evaluation cost with the actor and/or critic replaced by the oracle. Right: budget sweep (**budget_sweep_high_gap**, 2 seeds) — evaluation cost versus training episodes.

configuration	J (best iterate)			
		budget	markovian ₂	raw-history ₂
full learned (actor + critic)	2.82 ± 0.08	800	3.22	2.60
oracle critic, learned actor	2.56 ± 0.03	2400	3.22	2.46
oracle actor, learned critic	1.83 ± 0.00	7200	3.22	2.32

Replacing the actor cuts the cost most ($2.82 \rightarrow 1.83$), the critic less ($2.82 \rightarrow 2.56$): the residual is dominated by policy (gradient) extraction, consistent with the value/gradient gap of Proposition 6.4.

Raw-history keeps converging with budget; markovian is flat (representation-limited). Evidence that the learned H_1 failure on this cell is under-convergence (Finding 3.3).

- Comprehensive learned benchmark (Section 3, Table 2): **main_unified/** — per-cell sub-studies (***_study/summary.yaml**) sweeping capacity $\times 5$ seeds. Robustness (Appendix C): **main_unified/{oracle_ladder_high_gap, trick_ablation_convergence, budget_sweep_high_gap_b*}**.
- Canonical H_1/H_2 (5 cells $\times$ 5 seeds): **20260616_161650_on_off_sheet_array** (Table 3); on-sheet-only twin **20260616_161649_on_sheet_array** (Finding 4.5).
- Hopfield: **h1h2_tau_sweep_20260617_165104** (Table 4); **h1h2_tau2_control_20260617_171721**, **h1h2_tau2_ridge_20260617_180843**, **h1h2_tau3_control_20260617_194242**, **h1h2_tau3_offsheet_20260617_204438** (Table 5, Finding 5.3).
- Duffing: **h1h2_duffing_kappa_sweep_20260617_155613**; Mackey–Glass off-sheet: **h1h2_mg_offsheet_20260630_164840** (Table 6).
- Effective rank: **data/h1h2_effective_rank_vs_delay/** (Table 7).

Provenance caveats. The Hopfield/Mackey–Glass sweeps and the effective-rank measurement are single-seed (the operative quantity there is the data/conditioning axis, not seed variance); the canonical H_1/H_2 table is five-seed. The Mackey–Glass off-sheet curve is now complete for both cells (the last **mg_chaotic** point, off/dim = 4, at $\sim 10^4$ Pontryagin solves, finished after the first draft and is included). The trajectory figures are local regenerations whose well-conditioned costs match the cluster **summary.json** to four decimals; the ill-conditioned signature on the plateau

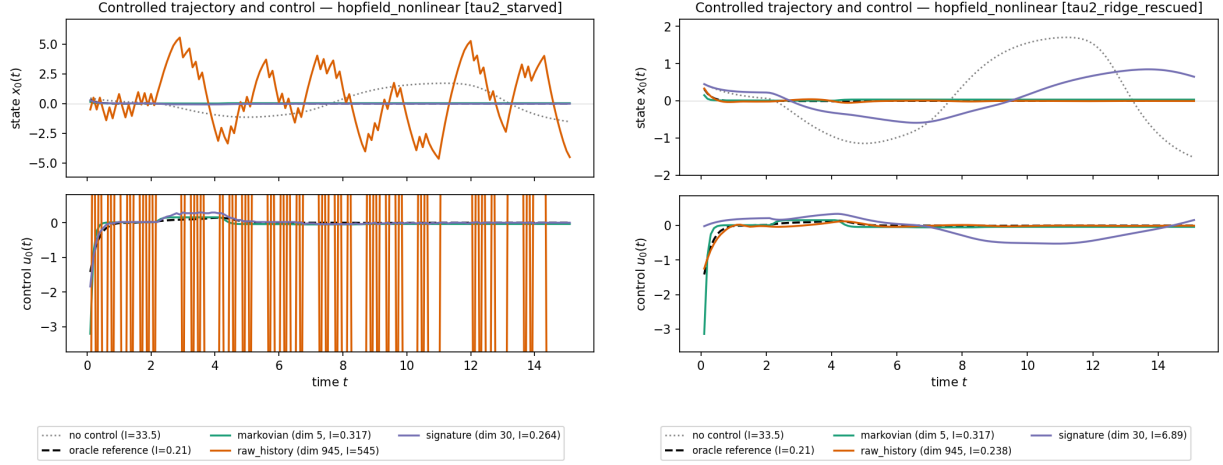


Figure 5: `hopfield_nonlinear` at $\tau = 2$: the H_2 illusion and its cure. Left, the starved fit ($n/\dim \phi_H = 4.4$): the history window (orange, dimension 945) diverges into a saturating bang-bang control ($I = 545$) while the signature (purple) and oracle (dashed) track, so the signature appears to win. Right, the same data with Tikhonov conditioning ($\lambda = 10^{-2}$): the history window is restored to $I = 0.238$ and tracks the oracle, while the identical ridge over-damps the signature ($I = 6.89$). The cure is conditioning, not capacity (Finding 5.2).

reproduces only to $\sim 10\%$ (local 0.140 versus cluster 0.164), itself a symptom of the condition number of Figure 3.

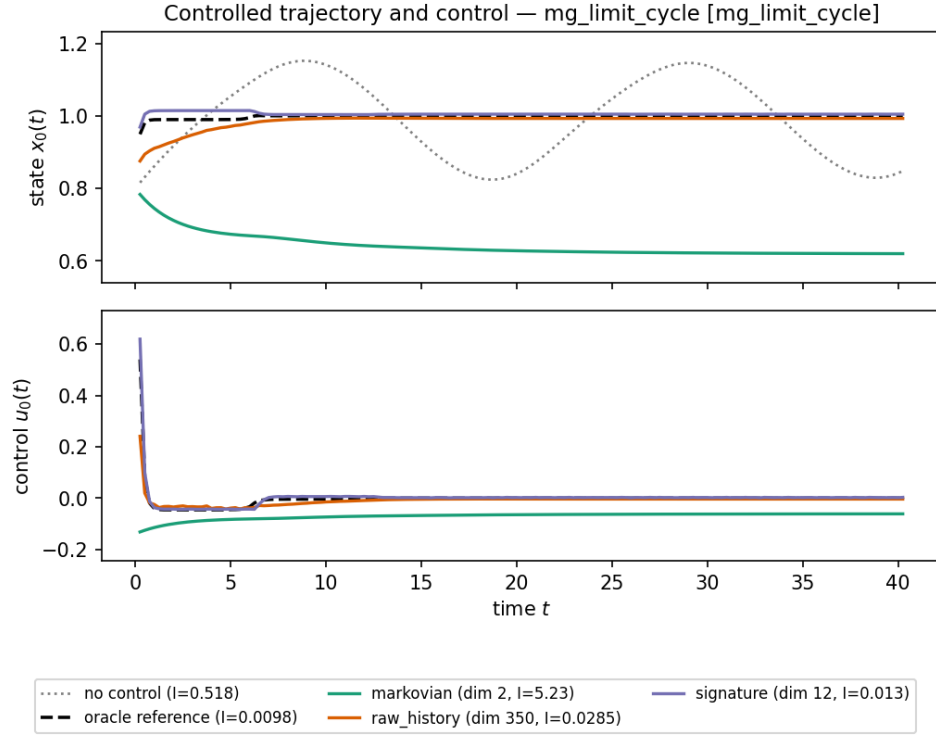


Figure 6: `mg_limit_cycle` (canonical off-sheet budget, $N_{\text{off}} = 500$). A clean H_1 picture: the markovian critic (green) cannot represent the value ($R^2 = 0.14$) and drives the state to a *wrong* equilibrium near $x_0 = 0.62$ (cost $I = 5.23$), while the history window (orange) and signature (purple) both regulate to $x^* = 1$. At this off-sheet budget the signature leads ($I = 0.013$ versus the history window’s 0.029); Finding 5.5 shows that scaling the off-sheet data to $\text{off}/\text{dim} = 4$ reverses the ordering.